ModelIt!

Student Activity Pack

The Mountain That Used to Be an Ocean

How Seashells End Up on Mountaintops Millions of Years Later

Grade 04L2 | NGSS 4-ESS1-1, 4-ESS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever found a seashell far from the ocean? Where do you think it came from?**

2. **How do you think mountains are formed? Do they grow, or are they pushed up?**

3. **Draw what you think the ground under a mountain looks like — are the layers flat or folded?**

4. **What would it mean to find a fossil of a fish on top of a mountain?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Plate Movement Speed | Sediment Deposit Rate | Compression Force | Rock Uplift | Fossil Elevation

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fast Collision, Thick Sediment

Plate Movement Speed at 80%, Sediment Deposit Rate at 80%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Slow Collision, Thin Sediment

Plate Movement Speed at 20%, Sediment Deposit Rate at 20%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: No Movement

Plate Movement Speed at 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Erosion Rate — How fast wind and water wear down the mountain after it forms — erosion can expose or destroy fossils

Rock Type — Whether the rock is hard or soft affects how much it deforms during compression — soft rock folds while hard rock cracks

Time Scale — How many millions of years the process has been happening — more time means more uplift and higher mountains

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How is it possible to find fossils of ocean creatures on top of mountains that are thousands of feet above sea level? Use evidence from your simulation to support your answers.

1. **How can finding a seashell on a mountaintop tell us about the history of that place?**

2. **Why do mountains like the Himalayas have fossils of ocean creatures but mountains like volcanic islands do not?**

3. **If tectonic plates suddenly stopped moving, would mountains keep growing? Why or why not?**

4. **How is squeezing play dough between your hands similar to what happens when tectonic plates collide?**

5. **How did your model help you understand something that takes millions of years to happen?**

8 STEM Challenge: Build a Tectonic Collision Model

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YOUR ENGINEERING MISSION

Design a hands-on model that demonstrates how tectonic plate collisions push ocean floor layers (with fossils) upward into mountains.

THE SCENARIO

A natural history museum wants a hands-on exhibit that shows visitors how ocean fossils end up on mountains. Your engineering team must build a physical model using layered materials that demonstrates how plate collision, compression, and uplift work together to push ocean floor sediment (with fossils) to high elevations.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE FOR ROCK LAYERS (CLAY, FOAM, TOWELS, CONSTRUCTION PAPER)?
2. HOW WILL YOU REPRESENT FOSSILS INSIDE THE LAYERS (SMALL SHELLS, BEADS, STICKERS)?
3. HOW WILL YOU SIMULATE TECTONIC PLATE COLLISION (PUSHING FROM BOTH SIDES)?
4. HOW WILL YOU SHOW VISITORS THAT THE FOSSILS MOVED FROM SEA LEVEL TO MOUNTAINTOP LEVEL?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geologists and Paleontologists study Earth's rocks, fossils, and history to understand how our planet has changed over billions of years. They discover ancient oceans, vanished continents, and extinct creatures. They earn $60,000–$110,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Scientists find fossils of ocean creatures on top of a mountain that is thousands of feet above sea level. How did those fossils get there?

- **A.** Ocean animals climbed up the mountain millions of years ago
- **B.** A giant wave carried the fossils to the mountaintop
- **C.** Tectonic plate collisions pushed ancient ocean floor rock upward over millions of years, carrying the fossils with it
- **D.** People carried the fossils up the mountain and left them there

2 In the model, Plate Movement Speed and Sediment Deposit Rate are external components. What does this mean?

- **A.** They are the newest parts of the model
- **B.** They are the starting conditions that drive the system, and the internal components respond to them
- **C.** They happen on the outside surface of the Earth
- **D.** They can be turned off to stop the model

3 The model shows that Plate Movement Speed has a POSITIVE relationship with Compression Force. What does this mean?

- **A.** Faster plate movement creates weaker compression because the plates slide past each other
- **B.** Plate Movement Speed and Compression Force are not related
- **C.** When Plate Movement Speed increases, Compression Force also increases because faster collisions push harder
- **D.** Compression Force stays the same no matter how fast the plates move

4 A student sets Plate Movement Speed to 0% in the model. What happens to all the internal components?

- **A.** Compression Force, Rock Uplift, and Fossil Elevation all continue increasing on their own
- **B.** Compression Force drops to zero, Rock Uplift stops, and Fossil Elevation stays where it is
- **C.** The fossils sink back down to the ocean floor
- **D.** Sediment Deposit Rate increases to make up for the missing plate movement

5 The model has two external components that both contribute to where fossils end up. Why do you need BOTH thick sediment layers AND strong plate movement to find many ocean fossils on mountaintops?

- **A.** Thick sediment makes the plates move faster
- **B.** Plate movement creates the fossils and sediment carries them upward
- **C.** Thick sediment layers contain the fossils, and plate movement creates the compression and uplift that pushes those layers to high elevations
- **D.** Neither one matters because fossils always end up at the same elevation